

SYSTEM ARCHITECTURE

PRESENTATION LAYER

Web App

Next.js 16 + React 19
Tailwind CSS 4 + shadcn/ui

Desktop App

Tauri v2 (Rust)
Windows NSIS / Linux deb

Progressive Web App

Service worker + manifest
Offline caching

UI Components

Design • Analysis • Results
30+ shadcn primitives

API LAYER

/api/predict

POST - AI Detection
Gemini / ONNX

/api/predict-local

POST - Local Models
Keras + YOLO

/api/design

POST - CTTT Engine
Reinforcement design

/api/export

POST - PDF Report
jsPDF generation

Engine

cttp-rules.ts
engine.ts

INFERENCE LAYER

Gemini 2.0 Flash

Cloud API — Distress detection

Python Inference Server

Flask — Port 5980

Keras

EfficientNetB0

YOLOv8

Classification

ONNX Runtime

CPU-only option

STORAGE & DEPLOYMENT

INFERENCE PROVIDERS ARE SWAPPABLE VIA ENVIRONMENT VARIABLES — ZERO CODE CHANGES